

CSC 6303 Final Exam Spring 1 2026

Important:

You have **four hours** to solve all eight questions.

You have to submit **one single pdf** with your answers. In this pdf you can put electronic edited text, photo of handwritten text or annotations, print screen images, or any other materials you believe to be appropriate as your answer.

Remember to number your answers from 1 to 8.

If you have any issues submitting your pdf file, please send an email immediately to **fernandes@merimack.edu** stating your issue and attaching your pdf file with answers.

1) Algol and C have both influenced most of the currently popular procedural languages. In your own words, what were the more important changes that C brought with respect to Algol?

Consider in your answer, technical aspects, not your personal preferences. Feel free to cite examples supporting your point.

Be concise, a short paragraph is generally enough to answer this question.

2) Convert from Python to C++

Convert the code below from Python to C++ assuming the use of streams *cin* and *cout* for input and output.

```
def RSP():
    guess = ""
    while (True):
        guess = input("Enter (r)ock, (s)cissors, or (p)aper: ")
        if (guess not in ["r", "s", "p"]):
            print("Only 'r', 's', or 'p' are valid inputs! Please try again.")
        else:
            break
    rand = random()
    if (rand < 1/3):
        comp = "r"
    elif (rand < 2/3):
        comp = "s"
    else:
        comp = "p"
    if (guess == comp):
        print("It is a tie!")
    elif ((guess == 'r') and (comp == 'p')) or \
          ((guess == 'p') and (comp == 's')) or \
          ((guess == 's') and (comp == 'r')):
        print("Sorry, you lost as I had", comp)
    else:
        print("Congrats, you won as I had", comp)

RSP()
```

3) Convert Python to Go

Convert the following Python function to Go:

```
def letter(score):  
    if score >= 90:  
        grade = "A"  
    elif score >= 80:  
        grade = "B"  
    elif score >= 70:  
        grade = "C"  
    else:  
        grade = "F"  
    if (score % 10 >= 7) and (grade not in ["A", "F"]):  
        grade += "+"  
    elif (score % 10 <= 3) and (grade != "F"):  
        grade += "-"  
    return grade
```

4) JavaScript to change an HTML file

The code below creates the web page as depicted, where when the user clicks on the buttons, the color of the text changes due to the JavaScript execution.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Button and Style</title>
    <style>
      h2 {color: red}
      body {background-color: beige; color: grey}
    </style>
  </head>
  <body>
    <h2>Click to change</h2>
    <p>Color and other properties can be changed using a JavaScript function.</p>
    <h3 id="change_js" style="color: green">This Text Changes Color</h3>
    <script>
      function changeColor(color) {
        if (color == 1) { document.getElementById("change_js").style.color = 'black'; }
        else if (color == 2) { document.getElementById("change_js").style.color = 'blue'; }
        else { document.getElementById("change_js").style.color = 'green'; } }
    </script>
    <button onclick="changeColor(1)">Black</button>
    <button onclick="changeColor(2)">Blue</button>
    <button onclick="changeColor(3)">Green</button>
  </body>
</html>
```

Click to change

Color and other properties can be changed using a JavaScript function.

This Text Changes Color

Black Blue Green

Edit this JavaScript function (HTML file available [here](#)) so the text content (innerText) also changes as the new examples (sequence of displays after button clicks):

Click to change

Color and other properties can be changed using a JavaScript function.

This text is Green

Black Blue Green

Click to change

Color and other properties can be changed using a JavaScript function.

This text is Black

Black Blue Green

Click to change

Color and other properties can be changed using a JavaScript function.

This text is Blue

Black Blue Green

5) bash script

The bash script code below is executed inside the folder `"/Users/pf/test"` which has the contents as demonstrated by the following printscreen

```
1  #!/bin/bash
2  cd ~/test
3  echo "Enter the folder to create"
4  read namedir
5  if [ -d "$namedir" ]
6  then
7  echo "Nothing to do, $namedir exists already"
8  else
9  `mkdir $namedir`
10 echo "Folder $namedir created!"
11 fi
```

```
[LAB-CS-MAC1:test pf$ pwd
/Users/pf/test
[LAB-CS-MAC1:test pf$ ls -lag
total 16
drwxr-xr-x  6 staff  192 Mar  2 13:46 .
drwxr-xr-x+ 57 staff 1824 Mar  2 13:35 ..
drwxr-xr-x  2 staff   64 Mar  2 13:43 fg
-rw-r--r--  1 staff  412 Mar  2 13:46 letter.py
-rwxr-xr-x  1 staff  193 Mar  2 13:42 my.sh
drwxr-xr-x  2 staff   64 Mar  2 13:35 my_files
LAB-CS-MAC1:test pf$
```

While running the script what happens if the users type it in:

- A. `"my_folder"`
- B. `"my_files"`
- C. `"letter.py"`

Give short answers to each item, add minimal explanation, if necessary.

6) Unix-like OS item permissions

What will the 10 characters permission descriptor of a file system item that is a directory that can be:

- read by all users;
- executed only by users belonging to the group of the item owner;
- written by the item owner alone.

7) Virtual Machines

What is the difference between a Virtual Machine app and a Virtual machine?

Be concise, a short paragraph is generally enough to answer this question.

8) **Multi Threads**

What is necessary for tasks within an algorithm to be able to be instantiated into multiple concurrent threads?

Be concise, a short paragraph is generally enough to answer this question.